

# Solar Panel Cleaning Robot — Complete Wiring Guide

Hardware: RPi 5 + RadioMaster RP3 V2 + NEO-M9N GPS + Pico 2 W + L298N

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## 1. RadioMaster RP3 V2 → Raspberry Pi 5

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| RP3 V2 Pin | Wire    | RPi 5 Pin | GPIO                    | Notes                                                        |
|------------|---------|-----------|-------------------------|--------------------------------------------------------------|
| TX         | →       | Pin 10    | GPIO 15                 | UART RX — MAVLink data in RX ←                               |
| Pin 8      | GPIO 14 | UART TX   | — optional telemetry 5V | ← Pin 2 5V Powers the receiver GND — Pin 6 GND Common ground |

Port : /dev/ttyAMA0 Baud : 460800 Run : python3 mavlink\_bridge.py

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## 2. SmartElex NEO-M9N GPS → Raspberry Pi 5

Option A — USB-UART Adapter (easiest, no config needed)

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NEO-M9N TX → USB-UART adapter TX pin → RPi 5 USB port NEO-M9N VCC → 3.3V from adapter NEO-M9N GND → Adapter GND Port: /dev/ttyUSB0

Option B — Direct GPIO Pins

| NEO-M9N Pin | Wire | RPi 5 Pin | GPIO   | Notes                   | TX | → | Pin 27 | GPIO 0 |
|-------------|------|-----------|--------|-------------------------|----|---|--------|--------|
| UART2 RX RX | ←    | Pin 28    | GPIO 1 | UART2 TX (optional) VCC | ←  |   | Pin 1  |        |
| 3.3V GND    | —    | Pin 6     | GND    |                         |    |   |        |        |

Enable UART2 — add to /boot/firmware/config.txt: dtoverlay=uart2

Port : /dev/ttyAMA2 Baud : 9600 (NEO-M9N default)

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### 3. Raspberry Pi 5 → Raspberry Pi Pico 2 W

| Pico 2W Pin | Wire | RPi 5 Pin | GPIO | Notes |
|-------------|------|-----------|------|-------|
|-------------|------|-----------|------|-------|

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|                     |   |        |        |                      |              |
|---------------------|---|--------|--------|----------------------|--------------|
| ———— GP4 (UART1 TX) | → | Pin 29 | GPIO 5 | RPi 5 receives data  | GP5 (UART1   |
| RX)                 | ← | Pin 7  | GPIO 4 | RPi 5 sends commands | GND          |
|                     |   |        |        |                      | — Pin 39 GND |

Common ground

Enable UART4 — add to /boot/firmware/config.txt: dtoverlay=uart4-pi5

Port : /dev/ttyAMA4 Baud : 115200 Run : python3 rc\_drive.py --pico /dev/ttyAMA4 --no-gps

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### 4. MPU-6050 IMU → Raspberry Pi Pico 2 W (I2C0)

| MPU-6050 Pin | Wire | Pico 2W Pin | GP | Notes |
|--------------|------|-------------|----|-------|
|--------------|------|-------------|----|-------|

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|           |   |        |         |                             |           |                                             |
|-----------|---|--------|---------|-----------------------------|-----------|---------------------------------------------|
| ———— VCC  | ← | Pin 36 | 3V3 OUT | Use 3.3V — NOT 5V           | GND       | — Pin 38                                    |
| GND SDA   | → | Pin 11 | GP8     | I2C0 data                   | SCL       | → Pin 12 GP9 I2C0                           |
| clock AD0 | — | GND    |         | Sets I2C address = 0x68     | INT       | → Pin 14                                    |
| GP10      |   |        |         | Optional interrupt (unused) | XDA / XCL | — not connected (auxiliary I2C, not needed) |

I2C address: 0x68 (AD0 → GND) 0x69 (AD0 → 3.3V) if you need two MPU-6050s on same bus

No pull-up resistors needed — the MPU-6050 breakout board includes them.

What the IMU provides:

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|                    |                                                      |
|--------------------|------------------------------------------------------|
| ———— Roll (+right) | — tilt detection when robot is on sloped solar panel |
|--------------------|------------------------------------------------------|

Pitch (+nose-up) — forward/backward tilt Yaw rate (gz) — integrated into relative heading (drifts, no compass) Safety cut-off — motors auto-stop if tilt exceeds 30°

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## 5. Raspberry Pi Pico 2 W → L298N Motor Driver

Pico 2W Pin      L298N Pin    Notes

|                           |                                       |                        |                               |           |     |
|---------------------------|---------------------------------------|------------------------|-------------------------------|-----------|-----|
| GP0                       | IN1                                   | Left motor direction A | GP1                           | IN2       |     |
| Left motor direction B    | GP2 (PWM)                             | ENA                    | Left motor speed control      | GP3       | IN3 |
| Right motor direction A   | GP6                                   | IN4                    | Right motor direction B       | GP7 (PWM) | ENB |
| Right motor speed control | GP26 (ADC)                            | —                      | Battery voltage (via divider) | GND       |     |
| GND                       | Common ground (tie all GNDs together) |                        |                               |           |     |

L298N Power Connections:

———— 12V battery + → L298N 12V terminal Battery GND → L298N GND (also connect to Pico GND) L298N 5V out → Pico VSYS (powers Pico from the battery) L298N Motor A → Left motor wires L298N Motor B → Right motor wires

Battery Voltage Divider (for GP26 ADC):

———— Battery + → R1 (100 kΩ) — GP26 R2 (33 kΩ) GND (Divider scales ~13V battery down to 3.3V ADC input)

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## 6. Full System Overview

RadioMaster TX —UART→ RPi 5 /dev/ttyAMA0 @ 460800 mavlink\_bridge.py NEO-M9N  
GPS —UART→ RPi 5 /dev/ttyUSB0 @ 9600 gps\_reader.py RPi 5 —UART→  
Pico 2W GP12/GP13 @ 115200 rc\_drive.py MPU-6050 IMU —I2C0→ Pico 2W  
GP8/GP9 @ 400kHz pico/main.py Pico 2W —GPIO→ L298N IN1-4 + ENA/B  
pico/main.py L298N —wire→ Left & Right DC motors 12V Battery —————→  
L298N → Pico VSYS (power supply)

## 7. RPi 5 config.txt — Required Changes (add before first boot)

File: /boot/firmware/config.txt

Add this line:

```
dtoverlay=uart4-pi5 # enables Pico link on GPIO12(TX)/GPIO13(RX) → /dev/ttyAMA4
```

GPS uses USB adapter (/dev/ttyUSB0) — no overlay needed.

Then reboot: `sudo reboot`

Verify UARTs appeared:

```
ls /dev/ttyAMA*
```

```
# Should show: /dev/ttyAMA0 /dev/ttyAMA4
```

## 8. Quick Start

Step 1 — Fix UART permissions (run once): `sudo bash fix_uart_permissions.sh`

Step 2 — Install Python dependencies: `pip install -r requirements_planner.txt`

Step 3 — Flash Pico 2 W with MicroPython, then copy: `pico/main.py` → Pico filesystem as `main.py`

Step 4 — Start the RC drive controller:

```
python3 rc_drive.py --no-gps
```

Step 5 — Open browser: `http://:5000`

Test connection only (no mission planner needed): `python3 test_connection.py`